

LESSON 5-2 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Area of Triangles**

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a triangle using one half the base times the height.

REMEMBER

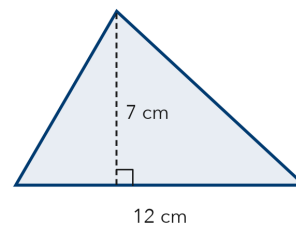
- A triangle is half of a parallelogram with the same base and height: $A = \frac{1}{2} \times b \times h$.
- The height meets the base at a right angle. It can fall **outside** the triangle.
- Area is in **square units** (cm², in², ft²).

WORKED EXAMPLE

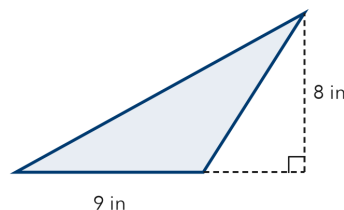
Find the area of a triangle with base 14 cm and height 9 cm.

Step 1 Write the formula: $A = \frac{1}{2} \times b \times h$ **Step 2** Substitute: $A = \frac{1}{2} \times 14 \times 9$ **Step 3** Multiply: $\frac{1}{2} \times 14 = 7$, and $7 \times 9 = 63$ **Answer** 63 cm²**WORD BANK****base** – the bottom side of the triangle**height** – straight up from the base, at a square corner $\frac{1}{2} \times b \times h$ – half of base times height**area** – space inside, in square units**PRACTICE***show your work · label units*

- 1.**
- Find the area of the triangle.

Step 1 Multiply the base and the height.**Step 2** Take half of that.*Show your work.***Answer** _____ cm²

- 2.**
- Find the area of the triangle. The height falls outside the triangle.

Step 1 Use 9 in and 8 in. The dashed line is the height.**Step 2** Multiply, then take half.*Show your work.***Answer** _____ in²

- 3.**
- Find the area of each triangle with base
- b
- and height
- h
- .

Step 1 Multiply $b \times h$ for each one.**Step 2** Take half.**a.** $b = 10$ cm, $h = 9$ cm _____ cm²**b.** $b = 5$ ft, $h = 11$ ft _____ ft²

4. **Garden Bed.** Maya's garden bed is a **triangle**: base 15 feet, height 6 feet. One bag of soil covers 9 square feet. How many bags does she need?

Step 1 Find the area: $\frac{1}{2} \times 15 \times 6$.

Step 2 Divide the area by 9.

Show your work.

Answer _____ bags

5. **Find the Error.** Jon says the area of a triangle with base 8 m and height 5 m is 40 m². Explain his mistake and find the correct area.

Frame Jon forgot to _____. The area is $\frac{1}{2} \times 8 \times 5 =$ _____ m².

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Find a triangle at home (a sign, a sandwich half, a roof). Estimate its base and height and its area.

LESSON 5-2 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2

Missing Measures in Triangles

Maryland Standard 6.G.A.1 • Learning target – I can find a missing base or height of a triangle when I know its area.

REMEMBER

- Know the area and one measure? **Double the area**, then divide by the measure you know: **$h = 2A \div b$** and **$b = 2A \div h$** .
- Check: $\frac{1}{2} \times b \times h$ should equal the given area.

WORKED EXAMPLE

Area = 30 in², base = 12 in. Find the height.

Step 1 Double the area: $2 \times 30 = 60$

Step 2 Divide by the base: $60 \div 12 = 5$

Step 3 Check: $\frac{1}{2} \times 12 \times 5 = 30$, so it checks

Answer **$h = 5$ in**

WORD BANK

missing measure – the base or height you need to find

double – multiply by 2

divide – share into equal groups; undo multiplying

check – put your answer back in $\frac{1}{2} \times b \times h$

PRACTICE

show your work • label units

1. Complete the table. Each row describes one triangle.

Step 1 Double the area.

Step 2 Divide by the measure you know.

	Base	Height	Area
Triangle A	8 in		28 in ²
Triangle B		9 ft	54 ft ²

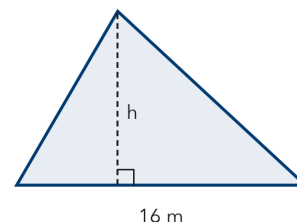
2. The area of the triangle is 48 m². Find the height h .

Step 1 Double the area: 2×48 .

Step 2 Divide by the base, 16.

Show your work.

Answer _____ m



3. Find the missing measure of each triangle.

a. $A = 40$ m², $b = 10$ m; $h =$ _____ m

b. $A = 27$ in², $h = 6$ in; $b =$ _____ in

4. **Sail.** A **sail** is a triangle. Its area is 35 square feet. Its height is 10 feet. How long is its base?

Step 1 Double the area: 2×35 .

Step 2 Divide by 10.

Answer _____ ft

5. **Compare.** Triangle P: base 12 cm, height 5 cm. Triangle Q: base 6 cm, height 10 cm. Which has more area? Explain.

Frame P has ____ cm^2 and Q has ____ cm^2 , so ____.

TEST PRACTICE

choose the best answer

6. A triangle has an area of 24 square units and a height of 4 units. What is the length of its base?

Step 1 Double 24.

Step 2 Divide by 4.

A 3 units

B 6 units

C 12 units

D 48 units

7. Which expression gives the area of a triangle with base 18 m and height 5 m?

A 18×5

B $\frac{1}{2} \times 18 \times 5$

C $18 + 5$

D $2 \times 18 \times 5$

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____